

Jack Riconosciuto

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SKILL SET

Programming Languages: C, C++, Scheme, Python, Assembly

Systems Programming: Strong background in low-level programming and performance optimization

Pure Math Background: graph theory, linear algebra, fourier analysis, automata theory

Optimization: Skilled in memory usage improvements, cache locality optimization, and holistic performance analysis beyond algorithmic complexity

Parallelism & Concurrency: Experienced in safe multithreading practices and efficient synchronization; knowledgeable in SIMD vectorization

Workflow Tools: Proficient with Valgrind (profiling and memory debugging), GDB (general debugging), the Godbolt Compiler Explorer (assembly/code generation analysis), and Git (version control)

COMPLETED PROJECTS

- [Kernel Space NVMe Driver](#) — *Spring 2025*
 - Wrote a custom driver for a proprietary research operating system at Northwestern. It is now in use and a part of the CS 343 curriculum.
- [Computational Genomics Project](#) — *Winter 2026*
 - I worked on a transcriptomics pipeline to run on Northwestern's Quest supercomputer to study differential gene expression in plants that perform facultative CAM photosynthesis.
- [Custom Network for Observatory](#) — *Fall 2026*
 - I created and currently maintain a computer system and local network for efficiently storing and processing all of the astrophotography club's photos.

JOBS

Computer Science Department — *Peer Mentor* September 2025 - Present

For Fall 25, Winter 26, and Fall 26, I have been a paid member of the teaching staff of CS 343 (Operating Systems), hosting office hours, helping plan assignments, and assisting the professor as needed. I also was a peer mentor for CS 446 in Spring 26, the graduate course on Kernel development.

Dearborn Observatory — *Guide* April 2024 - Present

Operate the telescope, maintain legacy computer systems and provide science communication during weekly public tours.

FNAL/BNL — *Intern* Summer 2024

Summer school experience at both Fermilab and Brookhaven National Labs, interacting with experts in particle physics, quantum computing, AI, and climate science.

Spanish Department — *Researcher* Summer 2026

After studying abroad in Barcelona in 2025, I contributed to a research project about Spanish Colonization funded by a grant I received from the Baker Program for Undergraduate Research. Responsibilities included compiling secondary sources, as well as transcribing 1700s manuscripts.

LANGUAGES

English, Spanish

EDUCATION

**Community High School,
West Chicago, IL**

Graduation: 5/2023

GPA: 5.541, Rank 3/420

**Northwestern University,
Evanston, IL**

Expected graduation: 3/2027

**Computer Science Major
Spanish Minor**

ACTIVITIES

Robotics

WeGo Robotics: *Team*

Captain/Founder

August 2021 - May 2023

Founded a FIRST Technical Challenge robotics team at my high school, winning the FIRST Rookie Award.

PWNAGE: *Safety Lead*

October 2020 - June 2022

FIRST Robotics PWNAGE team 2451, member of the electrical sub-team; in charge of ensuring the safety for the team. Qualified for the 2022 FIRST World Championships.

Astronomy Club

Outreach Lead/Founder

March 2026 - Present

Founded an Astronomy and Astrophotography club that utilizes the campus observatory. I am in charge of outreach, but also regularly assist the other leads in their tasks.

**Northwestern University
Marching Band**

Flute

August 2023 - Present

RELEVANT COURSEWORK

Operating Systems (CS 343)

Microcontroller System Design (CS 346)

Kernel Development (CS 446)

Networking (CS 340)

Data Structures/Algorithms (CS 214)

Programming Languages (CS 321)

Linear Algebra (Math 291)

Theory of Computation (CS 335)

Probability (Math 311)